

4.5 The substitution Rule.

$\int_a^b f(x) dx$ limit definition.

Fundamental theorem Calculus Pt. 2. $\longrightarrow$ We need an antiderivative for $f(x)$.

We want

$$\int 2x \sqrt{1+x^2} dx.$$

To find this integral we use the problem-solving strategy of introducing something extra. Here the "something extra" is a new variable.

Solution: Let $u = 1+x^2$. $du = (0+2x)dx = 2x dx$.

$$\int 2x \sqrt{1+x^2} dx = \int \sqrt{u} du = \int u^{1/2} du = \frac{2}{3} u^{3/2} + C$$

$$= \frac{2}{3} (1+x^2)^{3/2} + C.$$

$$\frac{d}{dx} \left\{ \frac{2}{3} (1+x^2)^{3/2} + C \right\} = \frac{2}{3} \cdot \frac{3}{2} (1+x^2)^{3/2-1} \cdot \frac{d}{dx} (1+x^2)$$

$$= (1+x^2)^{1/2} (2x)$$

$$= 2x \sqrt{1+x^2}.$$

In general, the method works whenever we have an integral that can be written in the form.

$$\int f(g(x)) \cdot g'(x) dx \quad \text{If } F' = f \text{ then}$$

$$\int F'(g(x)) g'(x) dx = F(g(x)) + C$$

If we make the "change variable" or "substitution" $u = g(x)$ then

$$\begin{aligned} \int F'(g(x)) g'(x) dx &= \int F'(u) du \quad \text{where } F' = f. \\ &= \int f(u) du. \end{aligned}$$

Thm. If $u=g(x)$ is a differentiable function whose range is an interval and f is continuous on I then.

$$\int f(g(x))g'(x)dx = \int f(u)du.$$

Example. Find $\int x^3 \cos(x^4+2)dx$.

Sln. $f(x) = \cos(x^4+2) = h(g(x))$ where $h(x) = \cos x$. $g(x) = x^4+2$

b/c $h(g(x)) = h(x^4+2) = \cos(x^4+2)$.

$g'(x) = 4x^3$. Let $u = x^4+2$ $du = 4x^3dx \Rightarrow x^3dx = \frac{du}{4}$.

$$\int x^3 \cos(x^4+2)dx = \int \cos(u) \frac{du}{4} = \frac{1}{4} \int \cos u du.$$

$$= \frac{1}{4} \sin u + C.$$

$$= \frac{1}{4} \sin(x^4+2) + C.$$

Problem: Evaluate $\int \sqrt{2x+1} dx$.

Sln.? $\int \sqrt{2x+1} dx = \int (2x+1)^{1/2} dx = \frac{(2x+1)^{1/2+1}}{1/2+1} + C = \frac{2}{3} (2x+1)^{3/2} + C.$

$$\frac{d}{dx} \left\{ \frac{2}{3} (2x+1)^{3/2} \right\} = \frac{2}{3} \cdot \frac{3}{2} \cdot (2x+1)^{3/2-1} \cdot \frac{d}{dx} (2x+1)$$

$$= \sqrt{2x+1} \cdot (2)$$

$$= 2\sqrt{2x+1}.$$

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C.$$

Example. Find $\int \frac{x}{\sqrt{1-4x^2}} dx$.

Sln: $\int \frac{x}{\sqrt{1-4x^2}} dx = \int x \cdot \frac{1}{\sqrt{1-4x^2}} dx.$

Let $f(x) = \frac{1}{\sqrt{1-4x^2}} = h(g(x))$ where $h(x) = \frac{1}{\sqrt{x}}$.

$g(x) = 1-4x^2 \Rightarrow g'(x) = -8x.$

Let $u = 1-4x^2$ $du = -8x dx$ then $x dx = -\frac{1}{8} du.$

$$\begin{aligned}
 \int \frac{x}{\sqrt{1-4x^2}} dx &= \int \frac{1}{\sqrt{u}} \left(-\frac{du}{8} \right) = -\frac{1}{8} \int \frac{1}{\sqrt{u}} du. \quad ; \quad x^{-n} = \frac{1}{x^n}. \\
 &= -\frac{1}{8} \int u^{-1/2} du \\
 &= -\frac{1}{8} \frac{u^{1/2}}{1/2} + C \\
 &= -\frac{1}{4} \sqrt{u} + C \\
 &= -\frac{1}{4} \sqrt{1-4x^2} + C.
 \end{aligned}$$

Example. Evaluate $\int \cos(5x) dx$.

Sln: $\int \cos(5x) dx = \int 1 \cdot \cos(5x) dx.$

Let $f(x) = \cos(5x) = h(g(x))$ where $h(x) = \cos x$ $g(x) = 5x$.
 $g'(x) = 5.$

Let $u = 5x$ $du = 5dx \Rightarrow \frac{du}{5} = dx.$

By subst.

$$\begin{aligned}
 \int \cos(5x) dx &= \int \cos u \frac{du}{5} = \frac{1}{5} \int \cos u du. = \frac{1}{5} \sin u + C. \\
 &= \frac{1}{5} \sin(5x) + C.
 \end{aligned}$$

Example. Find $\int \sqrt{1+x^2} x^5 dx.$

Sln. $f(x) = \sqrt{1+x^2} = h(g(x))$ where $h(x) = \sqrt{x}$ $g(x) = 1+x^2$.
 $g'(x) = 2x.$

Let $u = 1+x^2$ $du = 2x dx \Rightarrow x dx = \frac{1}{2} du.$

$$\begin{aligned}
 \int \sqrt{1+x^2} x^5 dx &= \int \sqrt{1+x^2} x^4 (x dx) \\
 &= \int \sqrt{u} x^4 \frac{1}{2} du.
 \end{aligned}$$

We need to find x^4 in terms of u . We know $u = 1+x^2 \Rightarrow x^2 = 1-u.$

then $x^4 = (1-u)^2.$

$$\begin{aligned}
\int \sqrt{1+x^2} \, x^4 (x dx) &= \int \sqrt{u} (1-u)^2 \frac{du}{2} \\
&= \frac{1}{2} \int \sqrt{u} (1-2u+u^2) du \\
&= \frac{1}{2} \int (u^{1/2} - 2u^{3/2} + u^{5/2}) du \\
&= \frac{1}{2} \left\{ \frac{2}{3} u^{3/2} - 2 \cdot \frac{2}{5} u^{5/2} + \frac{2}{7} u^{7/2} \right\} + C \\
&= \frac{1}{3} u^{3/2} - \frac{2}{5} u^{5/2} + \frac{1}{7} u^{7/2} + C \\
&= \frac{1}{3} (1+x^2)^{3/2} - \frac{2}{5} (1+x^2)^{5/2} + \frac{1}{7} (1+x^2)^{7/2} + C
\end{aligned}$$

Substitution: Definite integrals.

Evaluate $\int_0^4 \sqrt{2x+1} \, dx$.

Sln 1: Step 1: (find an antiderivative).

$$\begin{aligned}
&\int \sqrt{2x+1} \, dx \quad \text{let } u = 2x+1 \quad du = 2 \, dx \quad dx = \frac{1}{2} du \\
&= \frac{1}{2} \int \sqrt{u} \, du = \frac{1}{2} \cdot \frac{2}{3} u^{3/2} = \frac{1}{3} (2x+1)^{3/2}
\end{aligned}$$

Step 2: (Apply FTC P2):

$$\begin{aligned}
\int_0^4 \sqrt{2x+1} \, dx &= \frac{1}{3} (2x+1)^{3/2} \Big|_0^4 = \frac{1}{3} (8+1)^{3/2} - \frac{1}{3} = \frac{1}{3} (9)^{3/2} - \frac{1}{3} \\
&= \frac{1}{3} (27-1) = \frac{26}{3}.
\end{aligned}$$

Sln 2.

$$\begin{aligned}
\int_0^4 \sqrt{2x+1} \, dx &= \int_1^9 \sqrt{u} \frac{du}{2} \\
&= \frac{1}{2} \int_1^9 \sqrt{u} \, du \\
&= \frac{1}{2} \cdot \frac{2}{3} (u)^{3/2} \Big|_1^9 \\
&= \frac{1}{3} \{ 9^{3/2} - 1^{3/2} \} = \frac{1}{3} \{ 27 - 1 \} = \frac{26}{3}.
\end{aligned}$$

$$u = 2x+1$$

$$du = 2 \, dx \Rightarrow dx = \frac{1}{2} du$$

if $x=0$ then $u=1$

if $x=4$ then $u=9$

Thm. (The substitution rule for Definite integrals) If g' is continuous on $[a, b]$ and f is continuous on the range of $u = g(x)$ then

$$\int_a^b f(g(x))g'(x)dx = \int_{g(a)}^{g(b)} f(u)du.$$

Example. Evaluate $\int_1^2 \frac{dx}{(3-5x)^2}$.

Sln. Let $u = 3-5x$ $du = -5dx$ $dx = -\frac{1}{5}du$.

if $x=1$ then $u=-2$.

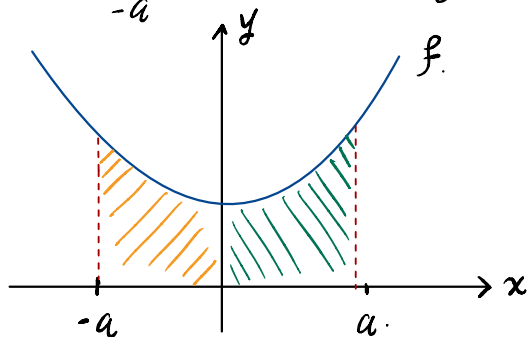
if $x=2$ then $u=-7$.

$$\begin{aligned} \int_1^2 \frac{dx}{(3-5x)^2} &= -\frac{1}{5} \int_{-2}^{-7} \frac{1}{u^2} du = -\frac{1}{5} \left(-\int_{-7}^{-2} u^{-2} du \right) \\ &= \frac{1}{5} \int_{-7}^{-2} u^{-2} du \\ &= -\frac{1}{5} \left. \frac{1}{u} \right|_{-7}^{-2} = -\frac{1}{5} \left(-\frac{1}{2} + \frac{1}{7} \right) = \frac{1}{14}. \end{aligned}$$

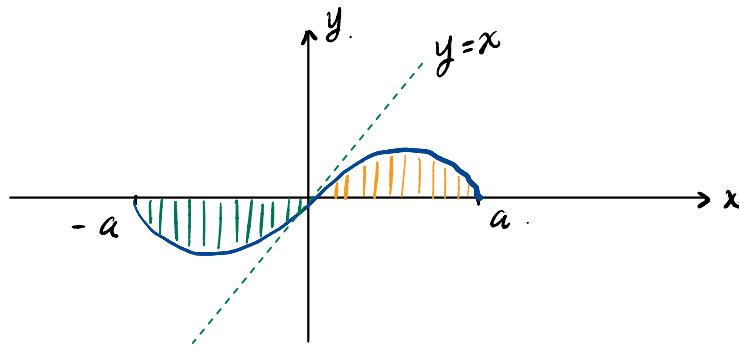
Symmetry: Recall that a function f is even if $f(-x) = f(x)$
 f is odd if $f(-x) = -f(x)$.

Thm: (Integrals of Symmetric functions) Suppose that f is continuous on $[-a, a]$.

a). If f is even then $\int_{-a}^a f(x)dx = 2 \int_0^a f(x)dx$.



b). If f is odd then $\int_{-a}^a f(x)dx = 0$.



Example. Because $f(x) = x^6 + 1$ satisfies $f(-x) = (-x)^6 + 1 = x^6 + 1 = f(x)$ it is an even function

$$\begin{aligned} \int_{-2}^2 (x^6 + 1) dx &= 2 \int_0^2 (x^6 + 1) dx = 2 \left\{ \frac{x^7}{7} + x \right\}_0^2 \\ &= 2 \left\{ \frac{2^7}{7} + 2 \right\} = \frac{284}{7}. \end{aligned}$$

Problem for you! Compute $\int_{-1}^1 \frac{\tan x}{1+x^2+x^4} dx$.